

20	A beam of light that consists of all wavelengths between 480 nm and 600 nm is projected onto a diffraction grating that contains 500 lines per millimeter. What is the maximum number of complete continuous spectra that can be observed emerging from the grating?										
	A	6		B	7		C	8		D	9

Answer: A

$$d \sin \theta = n\lambda$$

For longest wavelength, find the maximum order that can be observed:

$$n \lambda / d \leq 1 \rightarrow n \leq (1.0 \times 10^{-3} / 500) / 600 \times 10^{-9} = 3.3$$

$$\max n = 3$$

Thus there will be 6 spectra observed. (Zeroth order maximum is not a spectrum.)